

STD 1: Financial Maths (Std 1) F2 Investment (Y12)

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Exam Equivalent Time: 58.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

IMPORTANT FEATURES AND TIPS FROM 2UG EXAM HISTORY

- *MS-F2 Investments* has accounted for a very healthy 9% and 5% in the 2020 and 2019 Std1 exams respectively.
- *MS-F2 Investments* is a Year 12 Std1 topic that is comprised of the compound interest content found in Std2 and notably omits the content associated with shares and dividends.

ANALYSIS - What to Expect and Common pitfalls

- Examiners have tested the formula $FV = PV(1 + r)^n$ in each of the last two Std1 exams, including twice in 2020. This question type looks likely to consistently appear in the Std1 exam and any revision should reflect this.
- Testing compound interest through the *Compounded Value of \$1* tables has not been examined in the Std1 exam to date, but we note that it has been asked 5 times in the last decade in the Std2/Gen2 exam producing sub-50% mean marks on 4 occasions.
- Comparisons between compound and simple interest investments have appeared in each of the first two Std1 exams. This question type should be given a revision focus as we expect this trend to continue.
- All questions covering shares and dividends have been removed as they are beyond the scope of the Std1 course.

Questions

1. Financial Maths, STD2 F4 2009 HSC 6 MC

A house was purchased in 1984 for \$35 000. Assume that the value of the house has increased by 3% per annum since then.

Which expression gives the value of the house in 2009?

- (A) $35\,000(1 + 0.03)^{25}$
- (B) $35\,000(1 + 3)^{25}$
- (C) $35\,000 \times 25 \times 0.03$
- (D) $35\,000 \times 25 \times 3$

2. Financial Maths, STD2 F4 2015 HSC 17 MC

What amount must be invested now at 4% per annum, compounded quarterly, so that in five years it will have grown to \$60 000?

- (A) \$8919
- (B) \$11 156
- (C) \$49 173
- (D) \$49 316

3. Financial Maths, STD2 F4 2016 HSC 8 MC

The table shows the future value of an investment of \$1000, compounding yearly, at varying interest rates for different periods of time.

Future values of an investment of \$1000

Number of years	Interest rate per annum				
	1%	2%	3%	4%	5%
1	1010.00	1020.00	1030.00	1040.00	1050.00
2	1020.10	1040.40	1060.90	1081.60	1102.50
3	1030.30	1061.21	1092.73	1124.86	1157.63
4	1040.60	1082.43	1125.51	1169.86	1215.51
5	1051.01	1104.08	1159.27	1216.65	1276.28

Based on the information provided, what is the future value of an investment of \$2500 over 3 years at 4% pa?

- (A) \$1124.86
- (B) \$2812.15
- (C) \$3624.86
- (D) \$5312.15

4. Financial Maths, STD2 F4 2017 HSC 10 MC

A single amount of \$10 000 is invested for 4 years, earning interest at the rate of 3% per annum, compounded monthly.

Which expression will give the future value of the investment?

- A. $10\,000 \times (1 + 0.03)^4$
- B. $10\,000 \times (1 + 0.03)^{48}$
- C. $10\,000 \times \left(1 + \frac{0.03}{12}\right)^4$
- D. $10\,000 \times \left(1 + \frac{0.03}{12}\right)^{48}$

5. Financial Maths, STD2 F2 2021 HSC 5 MC

Peter currently earns \$21.50 per hour. His hourly wage will increase by 2.1% compounded each year for the next four years.

What will his hourly wage be after four years?

- A. $21.50(1.21)^4$
- B. $21.50(1.021)^4$
- C. $21.50 + 21.50 \times 0.21 \times 4$
- D. $21.50 + 21.50 \times 0.021 \times 4$

6. Financial Maths, STD2 F4 2012 HSC 9 MC

Tracy invests some money for 2 years at 4% per annum, compounded quarterly.

Compounded values of \$1

Period	Interest rate per period				
	1%	2%	3%	4%	5%
1	1.010	1.020	1.030	1.040	1.050
2	1.020	1.040	1.061	1.082	1.103
3	1.030	1.061	1.093	1.125	1.158
4	1.041	1.082	1.126	1.170	1.216
5	1.051	1.104	1.159	1.217	1.276
6	1.062	1.126	1.194	1.265	1.340
7	1.072	1.149	1.230	1.316	1.407
8	1.083	1.172	1.267	1.369	1.477

Which figure from the table should Tracy use to calculate the value of her investment at the end of 2 years?

- (A) 1.020
- (B) 1.082
- (C) 1.083
- (D) 1.369

7. Financial Maths, STD2 F4 2018 HSC 19 MC

The table shows the compounded values of \$1 at different interest rates over different periods.

Compounded values of \$1					
Number of periods	Interest rate per period				
	1%	2%	3%	4%	5%
2	1.0201	1.0404	1.0609	1.0816	1.1025
4	1.0406	1.0824	1.1255	1.1699	1.2155
6	1.0615	1.1262	1.1941	1.2653	1.3401
8	1.0829	1.1717	1.2668	1.3686	1.4775
10	1.1046	1.2190	1.3439	1.4802	1.6289
12	1.1268	1.2682	1.4258	1.6010	1.7959

Amy hopes to have \$21 000 in 2 years to buy a car. She opens an account today which pays interest of 4% pa, compounded quarterly.

Using the table, which expression calculates the minimum single sum that Amy needs to invest today to ensure she reaches her savings goal?

- A. $21\ 000 \times 1.0816$
- B. $21\ 000 \div 1.0816$
- C. $21\ 000 \times 1.0829$
- D. $21\ 000 \div 1.0829$

8. Financial Maths, STD1 F2 2020 HSC 8 MC

Joan invests \$200. She earns interest at 3% per annum, compounded monthly.

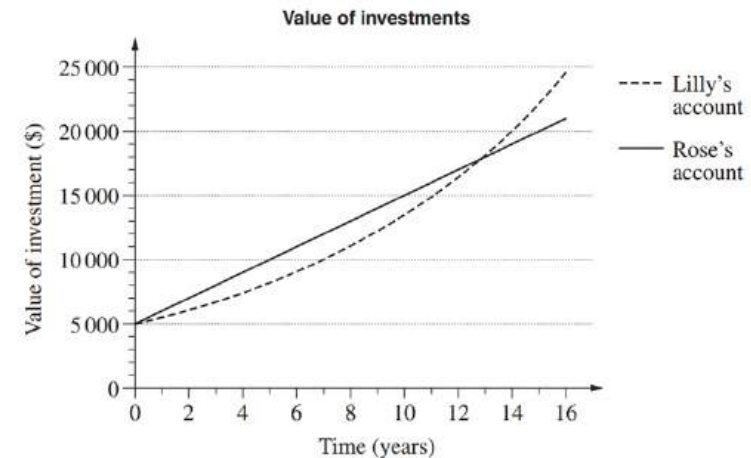
What is the future value of Joan's investment after 1.5 years?

- A. \$209.07
- B. \$209.19
- C. \$279.51
- D. \$311.93

9. Financial Maths, STD2 F1 2007 HSC 23a

Lilly and Rose each have money to invest and choose different investment accounts.

The graph shows the values of their investments over time.



- i. How much was Rose's original investment? (1 mark)
- ii. At the end of 6 years, which investment will be worth the most and by how much? (2 marks)
- iii. Lilly's investment will reach a value of \$20 000 first.
How much longer will it take Rose's investment to reach a value of \$20 000? (1 mark)

10. Financial Maths, STD2 F4 2008 HSC 24c

Heidi's funds in a superannuation scheme have a future value of \$740 000 in 20 years time. The interest rate is 4% per annum and earnings are calculated six-monthly.

What single amount could be invested now to produce the same result over the same period of time at the same interest rate? (3 marks)

11. Financial Maths, STD2 F4 2015 HSC 26d

A family currently pays \$320 for some groceries.

Assuming a constant annual inflation rate of 2.9%, calculate how much would be paid for the same groceries in 5 years' time. (2 marks)

12. Financial Maths, STD1 F2 2021 HSC 14

It costs \$2.45 for a car to travel on a toll road. Due to inflation, the cost is to increase by 3% each year.

How much will it cost for a car to travel on the toll road in 5 years time? (2 marks)

13. Financial Maths, STD2 F4 2020 HSC 21

The inflation rate over the year from January 2019 to January 2020 was 2%.

The cost of a school jumper in January 2020 was \$122.

Calculate the cost of the jumper in January 2019 assuming that the only change in the cost of the jumper was due to inflation. **(2 marks)**

14. Financial Maths, STD1 F2 2019 HSC 35

A bank offers two different savings accounts.

Account **X** offers simple interest of 7% per annum.

Account **Y** offers compound interest of 6% per annum compounded yearly.

The table displays the future values of \$20 000 invested in each account for the first 2 years.

<i>Future value</i>		
<i>End of year</i>	<i>Account X</i>	<i>Account Y</i>
1	\$21 400	\$21 200
2	\$22 800	\$22 472

- How much more money is there in Account **X** than in Account **Y** at the end of 2 years? **(1 mark)**
- Show that there would be more money in Account **Y** than in Account **X** at the end of 8 years. **(3 marks)**

15. Financial Maths, STD2 F4 2014 HSC 30a

Chandra and Sascha plan to have \$20 000 in an investment account in 15 years time for their grandchild's university fees.

The interest rate for the investment account will be fixed at 3% per annum compounded monthly.

Calculate the amount that they will need to deposit into the account now in order to achieve their plan. **(3 marks)**

16. Financial Maths, STD2 F4 2011 HSC 23c

An amount of \$5000 is invested at 10% per annum, compounded six-monthly.

Compounded values of \$1

Period	Interest rate per period				
	<i>1%</i>	<i>5%</i>	<i>10%</i>	<i>15%</i>	<i>20%</i>
1	1.010	1.050	1.100	1.150	1.200
2	1.020	1.103	1.210	1.323	1.440
3	1.030	1.158	1.331	1.521	1.728
4	1.041	1.216	1.464	1.750	2.074
5	1.051	1.276	1.611	2.011	2.488
6	1.062	1.340	1.772	2.313	2.986

Use the table to find the value of this investment at the end of three years. **(2 marks)**

17. Financial Maths, STD2 F4 2013 HSC 26e

Kimberley has invested \$3500.

Interest is compounded half-yearly at a rate of 2% per half-year.

Compounded values of \$1

<i>Period</i>	<i>Interest rate per period</i>					
	<i>1%</i>	<i>2%</i>	<i>3%</i>	<i>4%</i>	<i>5%</i>	<i>6%</i>
1	1.010	1.02	1.03	1.04	1.05	1.06
2	1.020	1.040	1.061	1.082	1.103	1.124
3	1.030	1.061	1.093	1.125	1.158	1.191
4	1.041	1.082	1.126	1.170	1.216	1.262
5	1.051	1.104	1.159	1.217	1.276	1.338
6	1.062	1.126	1.194	1.265	1.340	1.419
7	1.072	1.149	1.230	1.316	1.407	1.504
8	1.083	1.172	1.267	1.369	1.477	1.594

Use the table to calculate the value of her investment at the end of 4 years. **(2 marks)**

18. Financial Maths, STD1 F2 2020 HSC 13

Taro needs \$1000 in 5 years time. He is going to invest some money today in an account earning 3% per annum compounded annually. He will make no further deposits or withdrawals.

How much money does he need to invest today? **(3 marks)**

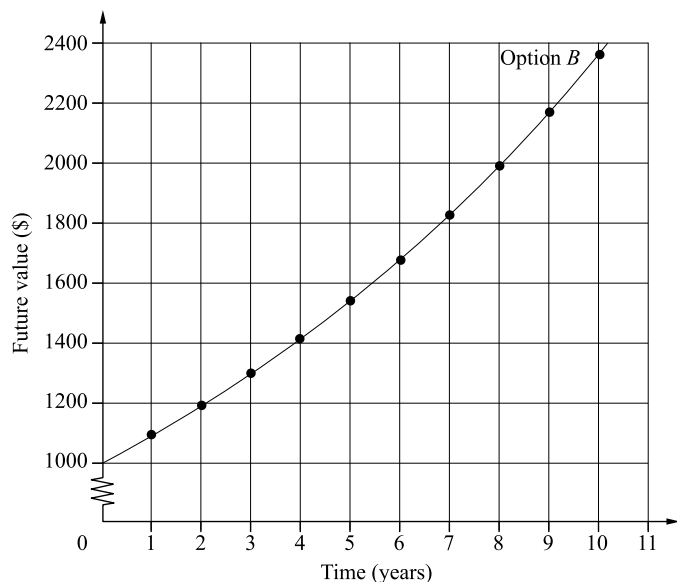
19. Financial Maths, STD1 F2 2020 HSC 25

Tom is offered two different investment options.

Option A: 10% per annum simple interest.

Option B: 9% per annum interest, compounded annually.

Tom has \$1000 to invest. The graph shows the future values over time of \$1000 invested using Option B.



Tom wants to find the difference between the future values after 8 years using these two investment options.

By first drawing, on the grid above, the graph of the future values of \$1000 invested using Option A, estimate the difference between the future values after 8 years. (3 marks)

20. Financial Maths, STD1 F2 2021 HSC 9 MC

James invests \$5000 at 3% per annum simple interest for 2 years, while Sally invests \$5000 at 3% per annum compounded annually for 2 years.

For the given investments, which of the following statements is TRUE?

- A. James earns more interest than Sally.
- B. Sally earns more interest than James.
- C. James and Sally earn the same amount of interest.
- D. There is not enough information to compare the interest earned by James and Sally.

Worked Solutions

1. Financial Maths, STD2 F4 2009 HSC 6 MC

$$r = 3\% = 0.03$$

$$n = 25 \text{ years}$$

$$\text{Using } FV = PV(1 + r)^n$$

$$\therefore \text{Value in 2009} = 35\,000(1 + 0.03)^{25}$$

$$\Rightarrow A$$

2. Financial Maths, STD2 F4 2015 HSC 17 MC

$$\text{Using } FV = PV(1 + r)^n$$

$$r = \frac{4\%}{4} = 1\% = 0.01 \text{ per quarter}$$

$$n = 5 \times 4 = 20 \text{ quarters}$$

$$60\,000 = PV(1 + 0.01)^{20}$$

$$\therefore PV = \frac{60\,000}{1.01^{20}}$$

$$= \$49\,172.66\dots$$

$$\Rightarrow C$$

3. Financial Maths, STD2 F4 2016 HSC 8 MC

$$\text{Table factor} = 1124.86$$

$$\therefore FV = 2.5 \times 1124.86$$

$$= \$2812.15$$

$$\Rightarrow B$$

4. Financial Maths, STD2 F4 2017 HSC 10 MC

$$\begin{aligned}\text{Compounding rate} &= \frac{3}{100} \div 12 \\ &= \frac{0.03}{12}\end{aligned}$$

$$\text{Compounding periods} = 4 \times 12 = 48$$

$$\therefore FV = 10\,000 \times \left(1 + \frac{0.03}{12}\right)^{48}$$

$\Rightarrow D$

5. Financial Maths, STD2 F2 2021 HSC 5 MC

$$\text{Wage after 1 year} = 21.50 \times 1.021$$

$$\text{Wage after 2 years} = 21.50 \times 1.021 \times 1.021 = 21.50(1.021)^2$$

$\vdots$

$$\text{Wage after 4 years} = 21.50(1.021)^4$$

$\Rightarrow B$

6. Financial Maths, STD2 F4 2012 HSC 9 MC

4% annual

$$\Rightarrow \frac{4\%}{4} = 1\% \text{ compounded each quarter}$$

$$\Rightarrow n = 8 \quad (8 \text{ quarters in 2 years})$$

$$\therefore \text{Factor} = 1.083 \quad (\text{from table})$$

$\Rightarrow C$

♦♦♦ Mean mark 24%. The lowest MC mean mark in the 2012 exam.
COMMENT: A process to follow: 1- convert the annual rate to the rate per compounding period. 2- calculate the number of compounding periods.

7. Financial Maths, STD2 F4 2018 HSC 19 MC

4% annual

$$\Rightarrow \frac{4\%}{4} = 1\% \text{ compounded quarterly}$$

$$\Rightarrow n = 8$$

$$\Rightarrow \text{Factor} = 1.0829$$

$$\therefore \text{Minimum sum} = 21\,000 \div 1.0829$$

$\Rightarrow D$

♦♦ Mean mark 33%.

8. Financial Maths, STD1 F2 2020 HSC 8 MC

$$\text{Monthly interest rate} = \frac{0.03}{12}$$

$$n = 1.5 \times 12 = 18$$

$$\begin{aligned}FV &= PV(1 + r)^n \\ &= 200 \left(1 + \frac{0.03}{12}\right)^{18} \\ &= \$209.19\end{aligned}$$

$\Rightarrow B$

♦♦ Mean mark 33%.

9. Financial Maths, STD2 F1 2007 HSC 23a

i. \$5000 (y -intercept)

ii. After 6 years,

$$\text{Lilly's investment} = \$9000$$

$$\text{Rose's investment} = \$11\,000$$

$$\therefore \text{Rose's is worth \$2000 more.}$$

iii. It takes Lilly 14 years to reach \$20 000 and it takes Rose 1 year longer (15 years) to reach the same value.

10. Financial Maths, STD2 F4 2008 HSC 24c

$$\begin{aligned}FV &= PV(1 + r)^n \\ 740\,000 &= PV \left(1 + \frac{2}{100}\right)^{40} \\ \therefore PV &= \frac{740\,000}{(1.02)^{40}} \\ &= 335\,138.907\dots \\ &= \$335\,138.91\end{aligned}$$

11. Financial Maths, STD2 F4 2015 HSC 26d

$$\begin{aligned}
 FV &= PV(1 + r)^n \\
 &= 320(1.029)^5 \\
 &= \$369.1703\dots \\
 &= \$369.17 \text{ (nearest cent)}
 \end{aligned}$$

12. Financial Maths, STD1 F2 2021 HSC 14

$$\begin{aligned}
 FV &= PV(1 + r)^n \\
 &= 2.45 \left(1 + \frac{3}{100}\right)^5 \\
 &= 2.45(1.03)^5 \\
 &= \$2.84
 \end{aligned}$$

13. Financial Maths, STD2 F4 2020 HSC 21

$$\begin{aligned}
 C_{2019} \times 1.02 &= 122 \\
 \therefore C_{2019} &= \frac{122}{1.02} \\
 &= \$199.61
 \end{aligned}$$

14. Financial Maths, STD1 F2 2019 HSC 35

a. Extra money in $X = 22\,800 - 22\,472$
 $= \$328$

b. Account X :

$$\begin{aligned}
 I &= Prn \\
 &= 20\,000 \times \frac{7}{100} \times 8 \\
 &= 11\,200
 \end{aligned}$$

$$\Rightarrow \text{Balance } X = 20\,000 + 11\,200 = \$31\,200$$

Account Y :

$$\begin{aligned}
 FV &= PV(1 + r)^n \\
 &= 20\,000 \left(1 + \frac{6}{100}\right)^8 \\
 &= \$31\,876.96
 \end{aligned}$$

$\therefore$ After 8 years, there's more money in Account Y .

♦ Mean mark part (b) 30%.

15. Financial Maths, STD2 F4 2014 HSC 30a

$$FV = \$20\,000, \quad n = 15 \times 12 = 180,$$

$$r = \frac{0.03}{12} = 0.0025$$

$$\begin{aligned}
 FV &= PV(1 + r)^n \\
 20\,000 &= PV(1 + 0.0025)^{180} \\
 PV &= \frac{20\,000}{(1.0025)^{180}} \\
 &= 12\,759.73\dots
 \end{aligned}$$

$\therefore$ They need to deposit \$12 760 (nearest \$)

♦ Mean mark 49%

16. Financial Maths, STD2 F4 2011 HSC 23c

Interest rate = 10% pa = 5% per 6 months

Period = 6 (6 x 6 months in 3 years)

⇒ Table value = 1.340

∴ Value of investment = 5000×1.34
= \$6700

♦♦ Mean mark 28%
MARKER'S COMMENT: Remember that the number of periods is the number of "compounding periods" and when asked to use the table, use the table!

17. Financial Maths, STD2 F4 2013 HSC 26e

$r = 2\%$ per half-year

$n = 8$ (8 half-years in 4 years)

⇒ Table Factor = 1.172

Investment = 3500×1.172
= \$4102

♦ Mean mark 44%
COMMENT: Structure your answer: 1-Find the interest rate per compounding period (same in this case). 2-Find the number of compounding periods.

18. Financial Maths, STD1 F2 2020 HSC 13

$$FV = PV(1 + r)^n$$

$$1000 = PV\left(1 + \frac{3}{100}\right)^5$$

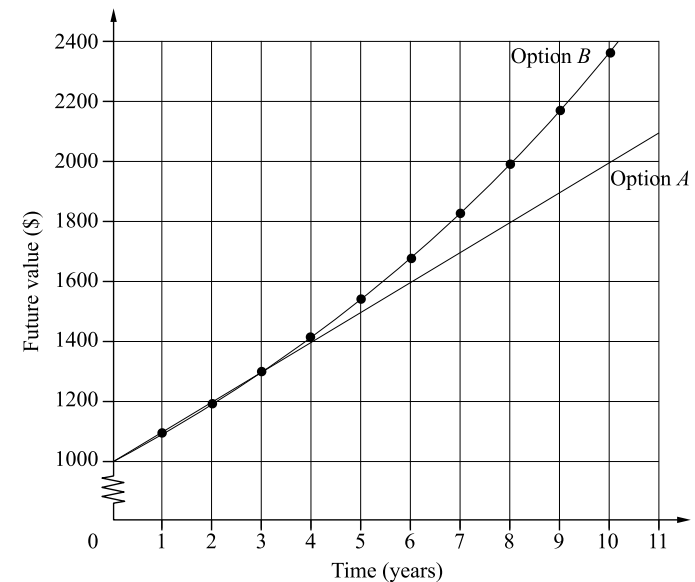
$$1000 = PV(1.03)^5$$

$$\therefore PV = \frac{1000}{(1.03)^5}$$

$$= \$862.61$$

♦♦ Mean mark 29%.

19. Financial Maths, STD1 F2 2020 HSC 25



When $t = 8$ years:

Option A = \$1800, Option B = \$2000

∴ Estimate of difference = $2000 - 1800$
= \$200

♦♦ Mean mark 30%.

20. Financial Maths, STD1 F2 2021 HSC 9 MC

Investments earn more when interest is compounded (vs simple interest) because interest is paid on both the principal and earlier interest.

∴ Sally earns more interest than James.

⇒ B